

AI in the SOC

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KEY CONCEPTS

- Foundations of AI, Machine Learning & Automation
- AI-powered threat detection & anomaly analysis
- AI in SIEM & EDR platforms
- Generative AI for investigation & reporting
- Analyst oversight, bias & responsible AI use

MAJOR TOPICS

- Anomaly detection & behavioral analytics
- AI-driven alerting & correlation in SIEM
- AI-based malicious behavior detection in EDR
- Using Generative AI for triage, investigations & reporting
- Managing false positives, bias & AI limitations



LEARNING OUTCOMES

- Explain how AI & ML are used in modern SOC's
- Interpret AI-generated alerts & risk scores
- Use Generative AI to support investigations & documentation
- Apply critical thinking to validate AI outputs

PREREQUISITES

- Basic understanding of SOC operations
- Familiarity with SIEM & EDR concepts

LET'S GO!



What Is Artificial Intelligence?



What Is AI?

“A machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations, or decisions influencing real or virtual environments.” - NIST

“An engineered system that generates outputs such as content, forecasts, recommendations or decisions for a given set of human-defined objectives.” - ISO/IEC 22989:2022

What is AI?

- Simulation of human intelligence in machines
 - Learning, reasoning, problem solving, language understanding, etc
- Narrow AI
 - Limited scope, non-sentient
- Artificial General Intelligence (AGI)
 - Theoretical concept, self-aware intelligence
- Machine learning (ML)
 - Algorithms that learn from large data sets
 - Make predictions or decisions, improve over time
 - Supervised
 - Unsupervised

Generative AI

- Algorithms that create content through generative modeling
 - Images, text, music, videos, etc
 - Matches or is similar to patterns and characteristics of training data
- Generative adversarial network (GAN)
 - Two neural networks used to train
 - Generator - creates new data
 - Discriminator - evaluates data from generator
 - Work against each other until discriminator unable to determine fake data

Large Language Models

- Type of generative AI trained on large amounts of data
 - Books, articles, blogs, social networks, etc
- Learns language patterns, grammar, syntax and context
- Specialize in mimicking content created by humans
- Text generation - predict next word in sentence based on context
- Common uses
 - Interactive conversation chatbots
 - Text-based content creation
 - Summarization of security log data

AI vs Machine Learning vs Automation



Why These Terms Get Confused

- + Vendors use terms inconsistently
- + Tools often combine all three
- + Analysts see outputs, not internals
- + Clarity improves trust and judgment

Automation: The Simplest Concept

- + Executes predefined actions
- + Follows explicit instructions
- + No learning or reasoning
- + Predictable and repeatable

Machine Learning

- Algorithms trained to make decisions or predictions
 - Without explicit programming
- Supervised learning
 - Trained on “labeled” data
 - Pictures of animals that are labeled (dogs, cats, horses) before training
- Unsupervised learning
 - Algorithm must find patterns & structures in unlabeled data
- Reinforcement learning
 - Training through trial and error with reward system

Artificial Intelligence: The Umbrella Term

Broad category of intelligent behavior

Includes rules,
ML, and
reasoning
systems

Produces
recommendations
or decisions

Often assists
humans, not
replaces them

Generative Artificial Intelligence



Gemini



How They Work Together in a SOC



How AI Is Used in Modern SOC's



AI in Cybersecurity

- AI and ML in cyber defense is not new
- Typical uses
 - Various automation
 - Threat detection and response
 - Anomaly detection
- Standalone AI solutions (generative AI, etc) are relatively new
- Volume of threats and sophistication have been increasing for years
 - Tools to combat are also evolving
- Augment and improve current processes
 - Not necessarily replace

Using AI

Behavior
Analytics



Advanced
Phishing
Detection



SIEM/SOAR



AI Goals

1

Augment and improve current processes and capabilities

- Detection of evolving threats

2

Make human-driven tasks easier, faster, more efficient

3

Reduce burden on analysts

Common AI and ML Terms You'll See in SOC Tools



Why These Terms Matter

- + SOC tools use AI/ML terminology in alerts
- + You don't need to build models
- + Recognition improves confidence
- + Terms explain *how* detections are made

Data and Training Terms

- + **Training data** – historical data used to learn patterns
- + **Baseline** – what ‘normal’ behavior looks like
- + **Features** – data points the model evaluates
- + **Telemetry** – raw data collected from systems

Model and Detection Terms

- + **Model** – the trained pattern recognizer
- + **Anomaly** – behavior that deviates from normal
- + **Classification** – labeling activity into categories
- + **Inference** – using a model to analyze new data

Scoring and Confidence Terms

- + Risk score – relative measure of concern
- + Confidence score – how sure the model is
- + Severity – prioritization level
- + Threshold – point where alerts trigger

SOC-Relevant AI Terms

- + UEBA – User and Entity Behavior Analytics
- + **False positive** – benign activity flagged as suspicious
- + **Drift** – behavior changes over time
- + **Human-in-the-loop** – analyst oversight

Generative AI

- + **Large Language Model (LLM)** - AI model trained on large amounts of text to generate and understand language
- + **Prompt** - The input or instruction given to the model
- + **Context** - Additional information provided to guide the response (alerts, logs, timelines)
- + **Token** - A chunk of text the model processes (words or parts of words)
- + **Completion / Response** - The output generated by the model
- + **Hallucination** - Confident-sounding but incorrect or unsupported output
- + **Guardrails** - Controls that limit what the model can see or do

Common AI and ML Terms You'll See in SOC Tools



Understanding Anomaly Detection at a High Level



Detection Approaches



Signatures are discrete. Anomalies are not.

What Anomaly Detection Is

- + Detects behavior that deviates from normal
- + Based on historical patterns
- + Often powered by machine learning
- + Focuses on “unusual,” not “malicious”

What Anomaly Detection Is Good At



New Threats

- + **Example: New malware or attack technique**
 - + A workstation begins making outbound connections to a domain that has **no known bad reputation** using legitimate software
- + **Why anomaly detection triggers:**
 - + This behavior **has never been seen before** in the environment
- + **SOC analyst takeaway:**
 - + “This isn’t known bad — but it’s new, rare, and unusual enough to deserve investigation.”

What Anomaly Detection Is Good At



New Threats



Behavior Changes

- + Example: Credential abuse starting quietly
 - + A user who normally logs in once per day now logs in **slightly more frequently** and login times gradually shift outside normal working hours
- + Why anomaly detection triggers:
 - + Over time, the **pattern** slowly drifts from the baseline
- + SOC analyst takeaway:
 - + “Nothing here breaks the rules — but the behavior is changing in a way attackers often start.”

What Anomaly Detection Is Good At



New Threats



Behavior Changes



Low and Slow Activity

- + Example: Lateral movement over days or weeks
 - + A compromised account accesses one new system per day and data transfer volumes are small and infrequent
- + Why anomaly detection triggers:
 - + The cumulative behavior is unusual compared to historical norms
- + SOC analyst takeaway:
 - + “Attackers are staying under the radar — anomaly detection catches what thresholds miss.”

Considerations for a SOC Analyst

- + Why Anomalies Create False Positives
 - + New software or workflows
 - + Admin or maintenance activity
 - + One-time or first-seen behavior
 - + Changing environments
- + How to Treat Anomalies
 - + Starting point, not a verdict
 - + Validate with logs and context
 - + Look for corroborating evidence
 - + Document conclusions clearly
- + Anomaly detection depends on a good baseline

Understanding Anomaly Detection at a High Level



How SIEMs Use AI for Alerting and Correlation



The Problem SIEMs Are Solving



Massive volumes of log and event data



Many signals are weak on their own



Manual correlation does not scale



Analysts need fewer, better alerts

Where AI Fits in SIEM Alerting

- + Detecting unusual or risky behavior
- + Scoring events and entities
- + Reducing alert noise
- + Prioritizing analyst attention

Considerations for a SOC Analyst

- + What Analysts See in Practice
 - + Fewer, higher-quality alerts
 - + Alerts tied to users or hosts
 - + Risk or confidence scores
 - + Contextual summaries
- + Analyst Responsibilities
 - + AI suggests, analysts confirm
 - + Correlation does not prove intent
 - + False positives still occur
 - + Evidence validation is required

Demonstration

How SIEMs Use AI for Alerting and Correlation



Interpreting AI-Generated Alerts and Risk Scores



What “AI-Generated Alerts” Really Mean

- + Key points:

- + AI in the SOC is primarily used to:
 - + Detect **anomalous behavior**
 - + Correlate weak signals across data sources
 - + Assign **risk or confidence scores**
- + These alerts are **not definitive incidents**
 - + They are *probabilistic assessments*, not facts
- + Most AI alerts answer:
“How unusual or risky is this compared to normal?”

What the Analyst Actually Sees

+ Common elements across tools:

- + Alert title or description
- + Entities involved (user, host, IP, process)
- + Time window and frequency
- + Supporting evidence or contributing signals
- + Risk score / severity / confidence

+ Examples:

- + Splunk: *Risk Score, Notable Event, Risk Object*
- + Elastic: *ML Anomaly Score, Anomaly Severity*
- + SIEMs generally: *High / Medium / Low + numeric score*

+ Key mindset shift:

Analysts investigate why the model is concerned, not just what happened.

Common False Positives and Why They Happen

+ Very common causes:

- + New software or patching activity
- + Admins behaving differently than usual
- + First-time logins, new devices, travel
- + Incomplete or noisy data sources

+ Why AI flags these:

- + Baselines haven't adapted yet
- + Models favor **sensitivity over certainty**
- + “Unusual” ≠ “Malicious”

+ Analyst responsibility:

- + Document false positives
- + Provide feedback (when tooling supports it)
- + Recognize recurring benign patterns

Key Takeaways for a Level 1 Analyst

+ Takeaways:

- + AI alerts are **starting points**, not conclusions
- + Risk scores help prioritize, not decide
- + Always validate with logs and context
- + Treat AI as a **junior analyst that never sleeps**

Interpreting AI-Generated Alerts and Risk Scores



How EDR Uses AI to Detect Malicious Behavior



What EDR Observes on the Endpoint

- + Continuous monitoring of endpoint activity
- + Process execution and parent-child relationships
- + Command-line arguments and script execution
- + File, registry, and memory behavior
- + Outbound network connections from processes

Why Behavior-Based Detection Is Required

- + Malware constantly changes hashes and filenames
- + Living-off-the-land tools appear legitimate
- + File reputation alone is unreliable
- + Attacker tradecraft is more consistent than malware

What an AI-Driven EDR Alert Looks Like

- + Detection name and mapped technique
- + Severity or confidence score
- + Affected endpoint and user
- + Timeline of actions
- + Visual process tree

AI in EDR Products

- + Microsoft Defender for Endpoint:
 - + Behavior-based detections
 - + Attack surface reduction + ML
- + Elastic Endpoint Security:
 - + ML-driven behavioral malware protection
 - + Anomaly detection on processes
- + CrowdStrike:
 - + Behavioral indicators of attack (IOAs)
 - + ML-based detection and prevention

Key Takeaways

- + EDR AI focuses on attacker behavior
- + Behavior detection works across malware variants
- + Process trees are critical investigation tools
- + Analysts confirm and contextualize detections

Interpreting AI-Generated Alerts and Risk Scores



Using AI to Assist Triage and Case Prioritization



The Triage Problem AI Helps Solve

- + High alert volume, limited analyst time
- + Not all alerts carry equal risk
- + Manual triage does not scale
- + AI helps surface the most important work

How AI Assists Triage and Prioritization

- + Risk scoring and confidence ranking
- + Behavioral context and baselining
- + Correlation across alerts and entities
- + Continuous reprioritization

What Analysts See in Practice

- + Ranked alert or case queues
- + High-risk entities highlighted
- + Contextual summaries of activity
- + Suggested next actions

AI Triaging in Tools

- + Splunk:
 - + Risk-Based Alerting (RBA)
 - + Entity risk scores
- + Elastic Security:
 - + Alert severity and ML anomaly scores
- + Microsoft Sentinel:
 - + Incident grouping and severity

Key Takeaways

- + AI reduces triage noise
- + Prioritization is dynamic, not static
- + Human judgment remains essential

Using AI to Assist Triage and Case Prioritization



How SOC's Use Generative AI



Follow Policy!



Generative AI: Quick Operational Review

- + Generates new content from learned patterns
- + Works with text, code, queries, and summaries
- + Best at language-heavy tasks
- + Augments analysts, does not replace detection engines

Common SOC Use Cases for Generative AI

- + Alert and case summarization
- + Log and event explanation
- + Query and search assistance
- + Knowledge base and runbook support

Generative AI in Triage and Investigations

- + Summarizes activity timelines
- + Highlights key entities and actions
- + Suggests investigative pivots
- + Reduces time-to-understand

Real-World Tool Examples

- + Microsoft Security Copilot:
 - + Incident summaries
 - + Query assistance
 - + Investigation guidance
- + Splunk:
 - + AI-assisted search and explanations
 - + SPL generation support
- + Elastic:
 - + Alert explanations
 - + Investigation summaries
- + Case management platforms:
 - + Report drafting
 - + Case notes and handoffs

How SOC's Use Generative AI



Writing Effective Prompts for Alert Triage



Why Prompts Matter in Alert Triage

- + GenAI responses depend heavily on input
- + Vague prompts produce vague answers
- + Good prompts save investigation time
- + Analysts remain responsible for outcomes

What Makes a Prompt Effective

- + Clear task or goal
- + Relevant context included
- + Specific questions asked
- + Appropriate scope

Poor vs Effective Prompt Examples

- + Poor: “Is this alert bad?”
- + Better: “Summarize the behavior that triggered this alert”
- + Poor: “Explain this log”
- + Better: “Explain why this process execution might be suspicious in a SOC context”

Prompts Analysts Can Use During Triage

- + “Summarize the key events in this alert”
- + “Explain why this behavior is unusual”
- + “What MITRE ATT&CK techniques might align with this activity?”
- + “What additional logs would help validate this alert?”

Guardrails and Best Practices

- + Never paste sensitive data unnecessarily
- + Validate AI output against logs
- + Treat responses as guidance, not fact
- + Document decisions in your own words

Key Takeaways for SOC Analysts

- + Prompts shape AI usefulness
- + Ask for explanations, not verdicts
- + Context improves results
- + Analysts remain accountable

Writing Effective Prompts for Alert Triage



Using GenAI to Assist Investigations



Where GenAI Fits in Investigations

- + Assists understanding, not detection
- + Works alongside SIEM and EDR tools
- + Best used after an alert is raised
- + Supports analyst decision-making

Common Investigation Tasks GenAI Supports

- + Summarizing alert timelines
- + Explaining logs and process behavior
- + Mapping activity to attack techniques
- + Identifying investigative pivots

Practical Prompt Examples for Investigations

- + “Summarize the sequence of events for this host”
- + “Explain why this process chain might indicate malicious behavior”
- + “Which MITRE ATT&CK techniques align with this activity?”
- + “What evidence would confirm or refute this hypothesis?”

Tool Examples in Real SOC Environments

- + Microsoft Security Copilot:
 - + Investigation summaries
 - + Threat technique mapping
- + Elastic AI Assistant:
 - + Alert explanations
 - + Query and pivot suggestions
- + Splunk AI features:
 - + Search assistance
 - + Log interpretation support

Guardrails: Using GenAI Safely

- + Always verify against raw evidence
- + Avoid sharing unnecessary sensitive data
- + Watch for confident but incorrect output
- + Keep humans in the decision loop

What This Means for SOC Analysts

- + Faster understanding of complex activity
- + Reduced cognitive load during investigations
- + Better documentation and handoffs
- + Analysts remain accountable

Demonstration

Demonstration

+ Alert text:

08/19/2018 22:24:47 PM LogName=Application SourceName=Symantec AntiVirus EventCode=51 EventType=2
Type=Error ComputerName=MKRAEUS-L.froth.ly TaskCategory=The operation completed successfully. OpCode=None
RecordNumber=7098 Keywords=Classic Message= Security Risk Found! Backdoor.PsEmpire in File:
C:\Users\MalloryKraeusen\AppData\Local\Packages\Microsoft.MicrosoftEdge_8wekyb3d8bbwe\AC\#!001\MicrosoftEdge\C
ache\PUZ8SSL4\BRUCE%20BIRTHDAY%20HAPPY%20HOUR%20PICS[1].lnk by: Auto-Protect scan. Action: Cleaned by
Deletion. Action Description: The file was deleted successfully.

Writing Effective Prompts for Alert Triage



Using GenAI for Reporting and Documentation



Where GenAI Fits in Documentation

- + Drafting incident summaries
- + Structuring case notes
- + Improving clarity and consistency
- + Rewriting technical details for different audiences

Practical Prompt Examples for Reporting

- + “Summarize this incident for a SOC handoff”
- + “Rewrite these findings in plain language”
- + “Create an executive-level incident summary”
- + “Structure these notes into a clear timeline”

SOC Tool Examples

- + Case management platforms:
 - + Drafting case summaries
 - + Analyst handoff notes
- + Microsoft Security Copilot:
 - + Incident summaries
 - + Executive reporting
- + SIEM integrations:
 - + Alert-to-report workflows

Guardrails and Best Practices

- + Analysts must verify all content
- + Never let AI invent facts
- + Avoid copying sensitive data blindly
- + Maintain ownership of reports

Key Takeaways for SOC Analysts

- + GenAI improves speed and consistency
- + Best used for drafting and rewriting
- + Accuracy and judgment remain human
- + Documentation quality matters

Data Summarization and Reporting

- Data pulled from multiple sources easily summarized
 - *“Look at the network and authentication logs from the past 7 hours. Tell me which what anomalies you see”*
- Easily highlight key findings, trends, vulnerabilities, etc
 - Pattern identification and highlighting
- Time savings through automation
 - Reduces MTTD (mean time to discovery)
 - Saves analyst time creating reports - only need to review/edit
- Reports easily tailored for different audiences

Demonstration

Using GenAI for Reporting and Documentation



AI Bias, False Positives, and Analyst Oversight



AI Is Powerful, Not Perfect

- + AI is trained on historical data
- + Models reflect data and design choices
- + AI outputs probabilities, not facts
- + Mistakes are expected, not failures

What Bias Looks Like in SOC AI

- + Biased or incomplete training data
- + Environments change faster than models
- + Over-representation of certain behaviors
- + Blind spots in visibility

Why False Positives Are Common

- + Unusual $\neq$ malicious
- + New software or workflows
- + Administrative activity
- + First-seen or rare behavior

Real Examples from SOC Tools

- + EDR flags admin scripts as malicious
- + SIEM anomaly alerts during maintenance
- + Identity alerts for legitimate travel
- + Baseline drift over time

The Analyst's Role in Oversight

- + Validate AI output with evidence
- + Provide context AI lacks
- + Recognize recurring false positives
- + Prevent blind trust or alert fatigue

Key Takeaways for SOC Analysts

- + AI bias and false positives are normal
- + Scores guide, not decide
- + Evidence always wins
- + Human oversight is essential

AI Bias, False Positives, and Analyst Oversight



How Custom ML Models Are Used in SOC's (High Level)



What “Custom ML Models” Mean in a SOC

- + Models built for a specific environment or problem
- + Trained on organization-specific data
- + Complement vendor-provided detections
- + Used when generic detections are not enough

Why SOCs Use Custom Models

- + Unique environments and workflows
- + High false positives from generic models
- + Organization-specific threat patterns
- + Need for tailored detection logic

Common Use Cases for Custom ML

- + User behavior anomaly detection
- + Service account misuse
- + Insider threat monitoring
- + Rare or low-and-slow attack patterns

What Analysts Should Expect

- + Alerts look similar to other AI detections
- + Scores and confidence levels still apply
- + False positives still occur
- + Analyst validation is required

How Custom ML Models Are Used in SOC's (High Level)



AI in the SOC - Summary

KEY CONCEPTS - RECAP

- **AI vs Machine Learning vs Automation**
- **AI-driven anomaly detection & behavioral analytics**
- **AI integration in SIEM & EDR**
- **Generative AI in SOC workflows**
- **Human oversight & responsible AI use**



LEARNING OUTCOMES - RECAP

- Explain AI applications in SOC environments
- Interpret & validate AI-generated alerts
- Use Generative AI to assist investigations & reporting
- Recognize AI limitations & bias

REAL-WORLD APPLICATIONS

- Accelerating alert triage in high-volume SOC environments
- Supporting investigations with AI-assisted log analysis
- Improve reporting efficiency with Generative AI tools

NEXT STEPS

- Practice reviewing AI-generated alerts in lab environments
- Explore advanced detection engineering concepts
- Continue learning about AI model tuning & SOC optimization

THANKS FOR WATCHING!



EXPERTS AT MAKING YOU AN EXPERT

